A Droop Quota DAE Model for *Mycobacterium smegmatis* Batch Cultivation Under Nutrient-Replete and Nitrogen-Limited Conditions

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***Abstract*—The nonpathogenic *Mycobacterium smegmatis* is a common surrogate of *M. tuberculosis*, which survives in nutrient-limited niches. We extend a differential-algebraic equation (DAE) model of *M. smegmatis* batch cultivation calibrated in nutrient-replete conditions (baseline model) with an internal nitrogen quota, a Han and Levenspiel-type biomass self-inhibition, a bounded pH factor, biomass-dependent carbon maintenance and ammonium in the charge balance (extended model). The extended model was calibrated on ammonium-deficient (nitrogen-limited) data only, using 5 of its 21 parameters (chosen among nested calibration subsets by information criteria, *F*-tests and identifiability); other datasets are tests without recalibration. Global normalized root-mean-square errors (NRMSE, percent of the data range) under ammonium deficiency fell from 47.4 to 5.1 percent (calibration) and from 41.2 to 19.6 percent (validation). Under nutrient-replete conditions, where the extended model was not calibrated, they rose from 18.5 to 20.9 percent (calibration) and from 12.3 to 15.5 percent (validation): a trade-off, not an improvement in every condition. All calibrated parameters had *t*-values above 4, but the Droop factor stays at or above 0.998: the quota is a nitrogen pool diluted by growth and the capacity term sets the biomass plateau. The results support the extended model as a whole, not individual parameter values or mechanisms.**

***Index Terms*—differential-algebraic equations, Droop kinetics, *Mycobacterium smegmatis*, parameter calibration, identifiability, tuberculosis.**

I. Introduction

Tuberculosis (TB) is the leading cause of death from a single infectious agent (an estimated 10.7 million cases in 2024 [1]), and its agent, *Mycobacterium tuberculosis*, survives in macrophages and granulomas, where hypoxia, nutrient limitation and low pH promote dormancy and persistence [2]. Because work with it requires high-containment facilities [3], the nonpathogenic, fast-growing *Mycobacterium smegmatis* is used as a surrogate [4], and mathematical models of its behavior under stress can complement experiments by exploring conditions that are costly to test.

In *M. smegmatis*, nitrogen limitation triggers a dedicated regulatory response [5], intracellular glutamine rises with nitrogen availability [6], and glycerol uptake has been reported to continue after growth stops [7]. Apiyo *et al.* [3] modeled an internal nitrogen storage whose uptake shuts off as the pool fills. In contrast, a fixed biomass-per-nitrogen yield ties nitrogen consumption rigidly to growth, and this is the assumption we relax.

This work builds on the differential-algebraic equation (DAE) framework of Apiyo *et al.* [3], extended by Díaz-Figueroa *et al.* [8] and de Witt *et al.* [9], which couple the culture environment, including pH, with biomass growth and substrate consumption. The model of [8], hereafter the *baseline model*, was calibrated only on nutrient-replete data. We evaluate it under severe ammonium deficiency, a condition for which it was not calibrated (the "nitrogen-limited" of the title refers to this imposed condition, not to a growth limitation in the calibrated model), and reformulate its nitrogen balance around a Droop-type internal quota that decouples uptake from growth. Because the model of [3] already included

an internal nitrogen storage, the novelty is relative to the baseline model. The resulting *extended model*, calibrated on ammonium-deficient data only and evaluated on all other datasets, also modifies self-inhibition, replaces the pH factor by a bounded one, and adds biomass-dependent maintenance and ammonium in the charge balance. This paper contributes the extended model, an evaluation of the baseline model under ammonium deficiency, and a comparison of both models on separate calibration and validation datasets of both conditions with a parameter reduction by nested comparison; together they assess a trade-off under limited data, not a validated description of *M. smegmatis* nor a claim that the extended model is better in every condition.

II. Methodology

A. Baseline and Extended Models

The baseline model is the oxygen-augmented DAE model of [8] with Monod growth kinetics: five differential states (X, C, N, CO_2, O_2) and the algebraic pH, obtained from an electroneutrality constraint without ammonium (Supplementary Material, Sect. S1.1). Its growth rate μ_D has a lag term, Monod terms for glycerol, oxygen and external ammonium ($N/(N+k_N)$), a logistic capacity term $1 - X/X_{\max}$ and a pH factor that, as implemented in the code of [8], is greater than or equal to one rather than a bounded inhibition factor. Nitrogen is consumed in fixed proportion to the biomass formed, $dN/dt = -\mu_D X/Y_{X/N}$, with a constant yield $Y_{X/N}$ (g X/g N).

The extended model adds a state for the internal nitrogen quota q (g N/g X) based on Droop kinetics [10], [11]. It is mechanistically motivated and structurally borrowed from established frameworks, not empirically validated for *M. smegmatis*. The complete system has six differential states (X, C, N, CO_2, O_2 and q) and one algebraic variable, the pH, defined implicitly by the constraint (7):

$$\frac{dX}{dt} = (\mu - k_d)\,X, \tag{1}$$

$$\frac{dC}{dt} = -\left(\frac{\mu}{Y_{X/C}} + m(X)\right) X, \tag{2}$$

$$\frac{dN}{dt} = -\rho\,X + f_{Nrec}\,k_d\,q\,X, \tag{3}$$

$$\frac{dCO_2}{dt} = \left(\frac{\mu}{Y_{X/CO_2}} + m(X)\,Y^m_{CO_2}\right) X - k_{a7}\frac{CO_2}{10^{-pH}/k_{a7} + 1}, \tag{4}$$

$$\frac{dO_2}{dt} = k_L a\,(O_{sat} - O_2) - \left(\frac{\mu}{Y_{X/O_2}} + m(X)\,Y^m_{O_2}\right) X, \tag{5}$$

$$\frac{dq}{dt} = \rho - \mu\,q, \tag{6}$$

$$0 = \mathcal{G}(pH, CO_2, N). \tag{7}$$

Here $\mathcal{G}$ is the electroneutrality constraint (below), $k_{a7} = 10^{-pK_{a7}}$ the CO_2/HCO_3^- dissociation constant, k_d the death-rate constant and f_{Nrec} the fraction of the quota of dying cells that returns to the medium. The specific growth, nitrogen uptake and maintenance rates are

$$\mu = \mu_{\max}\,f_{lag}\,f_C\,f_O\,f_{cap}\,I_{pH}\,f_{Nq}, \tag{8}$$

$$\rho = \rho_{N,\max}\,\frac{N}{N + k_{Nup}} \cdot \max\left(\frac{q_{\max} - q}{q_{\max} - q_{\min}}, 0\right), \tag{9}$$

$$m(X) = m_C\left(1 - \frac{X}{X_{\max}}\right)^{m_{cap}}, \tag{10}$$

with the lag and Monod factors $f_{lag} = 1 - e^{-t/t_{lag}}$, $f_C = C/(C + k_C)$ and $f_O = O_2/(O_2 + k_O)$, and

$$f_{cap} = \left(1 - \frac{X}{X_{\max}}\right)^{n_{cap}}, \quad I_{pH} = \exp\left[-\frac{(pH - pH_{opt})^2}{pH_w^2}\right].$$

Unlike the pH factor of the baseline code [8], which is ≥ 1, I_{pH} is a dimensionless inhibition factor with $0 < I_{pH} \leq 1$ (equal to 1 at pH_{opt}). The parameters $q_{\min}$ and $q_{\max}$ bound the quota. $q_{\min}$ is the minimum (subsistence) quota of Droop kinetics: the Droop factor $f_{Nq} = \max(1 - q_{\min}/q, 0)$ stops growth when q falls to $q_{\min}$. $q_{\max}$ is the maximum quota: the feedback term of (9) shuts uptake off as q approaches $q_{\max}$, and $q_{\max} \to \infty$ recovers uptake without feedback [3], [11]. Both are assumed values, not calibrated: $q_{\min} = 10^{-4}$ g N/g X and $q_{\max} = q(0)$, the initial quota. f_{Nq} replaces the external-nitrogen Monod term of the baseline model, and (3) and (6) replace its fixed-yield balance.

Monod terms for glycerol and oxygen follow [12]. The factor f_{cap} is a self-inhibition term in the style of Han and Levenspiel [13] with critical biomass $X_{\max}$ (without their scaling of the Monod constants); the exponent n_{cap} lets $X_{\max}$ be estimated separately from the growth dynamics.

Glycerol consumption has a growth part (yield $Y_{X/C}$) and a maintenance part [14]. Because a constant coefficient keeps consuming glycerol at high cell density, $m(X)$ decays with biomass from m_C at $X = 0$, with exponent m_{cap}, a phenomenological choice without a glycerol saturation term, motivated (not validated) by the adaptation of mycobacterial energy metabolism to carbon limitation and hypoxia [15]. The CO_2 and O_2 balances use the same fluxes, with yields Y_{X/CO_2}, Y_{X/O_2}, maintenance coefficients $Y^m_{CO_2}$, $Y^m_{O_2}$ and gas–liquid exchange.

The pH follows at every instant from an electroneutrality constraint over the phosphate, citrate, ammonium and carbonate buffers, based on [9]. The main change with respect to the baseline is that ammonium enters the charge balance, with $[NH_4^+]$ computed from N through its dissociation equilibrium, so that nitrogen assimilation acidifies the medium; the charge of every buffer increases monotonically with pH, so the solution is unique (Supplementary Material, Sect. S1.3).

The model is an index-1 DAE with 21 parameters (Table I); the acid–base constants, medium concentrations, pH_{alk}, $k_L a$ and O_{sat} are shared by both models, taken from the baseline code of [8] and traced to [16] (Table S2).

B. Experimental Data and Evaluation Protocol

The data were provided by Apiyo *et al.* [3]: batch cultures on glycerol with measured X, C, N and pH. Each condition has one calibration dataset (10 samples, 0–40 h) and one

independent validation dataset (10 samples, 0–99.5 h); initial ammonium nitrogen was about 1 g N/L in the nutrient-replete and 0.03–0.04 g N/L (roughly 30-fold lower) in the ammonium-deficient datasets. Initial X, C, N and pH are the first sample of each dataset; CO_2 and O_2 were not measured, and their initial values and $q(0) = 0.096$ g N/g X (the nitrogen fraction of the biomass formula of [16], not measured) are set as in Sect. S2 of the Supplementary Material.

Each model was calibrated on a single calibration dataset (extended: ammonium deficient; baseline: nutrient replete); all other datasets are tests without refit. The baseline is the model of [8] with its published calibrated parameters ($Y_{X/N}$, $X_{\max}$, $\mu_{\max}$), not recalibrated; its errors were recomputed with the metrics used here so that both models are compared on the same basis.

C. Metrics, Identifiability and Model Comparison

For variable $v \in \{X, C, N, pH\}$ with measurements $y_{v,i}$, simulations $\hat{y}_{v,i}$ at the $n_t = 10$ sampling times, residuals $e_{v,i} = y_{v,i} - \hat{y}_{v,i}$ and range R_v of the measured values, the normalized root-mean-square error is $\mathrm{NRMSE}_v = R_v^{-1}\sqrt{n_t^{-1}\sum_i e_{v,i}^2}$. The *global NRMSE* is $\sqrt{\widetilde{RSS}/n}$, with the range-scaled residual sum of squares $\widetilde{RSS} = \sum_v \sum_i (e_{v,i}/R_v)^2$ over the four variables and $n = 4n_t = 40$, the root mean square of the four per-variable values; $\widetilde{RSS}$ is also the calibration cost. Errors are in % of the range of the measured values in the dataset being evaluated.

Sensitivity and identifiability. Relative sensitivities of X, C, N and pH to each parameter were computed by central differences ($\pm 5\%$) at the calibration sampling times, a parameter below 0.01 for every variable being nonsensitive (Supplementary Material, Sect. S7.1). Practical identifiability was assessed at the calibrated parameters of each model, on its own calibration dataset, with the Fisher information matrix (FIM) of the measured variables (Sect. S7.2): the t-value is $t_j = |\theta_j|/\mathrm{SE}_j$ (significant if $t_j > 2$), with SE_j from the pseudo-inverse of the FIM, and two estimates are highly correlated if $|\kappa_{jk}| > 0.95$; both are linearized, pragmatic screens (also used in [17]).

Model comparison. Nested calibration subsets of the extended model were compared with an F-test of the reduction of $\widetilde{RSS}$ and the Akaike information criterion with small-sample correction (AICc) [18] for Gaussian errors with unknown variance; ΔAICc is the value of the larger subset minus that of the smaller, and Akaike weights are the normalized $e^{-\Delta/2}$. AIC, BIC and all formulas are in the Supplementary Material (Sect. S4.1). These criteria assume independent Gaussian residuals, which time series do not satisfy, so they rank models and are not tests; the two-unit tie band is a convention.

D. Calibration and Model Reduction

Parameters were calibrated by particle swarm optimization (PSO, `mealpy`, 50 particles, 200 epochs, $c_1 = c_2 = 1.5$, inertia 0.5, in every series), minimizing $\widetilde{RSS}$ on the calibration dataset, with simulation in CasADi/IDAS using the package of [19]. A *series* refers to a set of 10 independent PSO runs of one calibration subset, differing only in the random seed. In all, 686 runs were done in 59 series, in two stages. The first stage (552 runs, 42 series), began with 10–12 free parameters and wide bounds, or testing parameters one at a time. Overall, parameters on a bound, with $t < 2$ or strongly correlated with others were fixed or had their bounds changed (by hand, as in [20]), so later bounds depended on earlier results.

To remove that dependence, a final stage (134 runs, 17 series) fixed in advance the bounds of the five parameters of the selected set ($\mu_{\max}$, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, n_{cap}; Supplementary Material, Sect. S3.5) and the assumed values of the rest, added each fixed parameter to that pool (13 series) and compared the nested calibration subsets with admissible runs. A run was *admissible* if no calibrated parameter ended on a bound and all t-values were above 2. The reported set is the best of the ten runs of the chosen subset. Two exploratory recalibrations calibrated $q_{\min}$ or used constant maintenance, $m(X) = m_C$ (Supplementary Material, Sect. S6.2).

The decision rule used only the ammonium-deficient calibration dataset (error, information criteria, F-tests, FIM). The other datasets were not part of the objective and were used only to compare, not to decide, but their errors were computed for every run and seen alongside the criteria, so the final comparison is not blind to them. The fixed parameters keep the values of [8] or assumed values.

TABLE I
PARAMETERS OF THE EXTENDED MODEL AND THEIR VALUES IN THE BASELINE MODEL.

Parameter	Units	Baseline		Extended	
		Value	t-value	Value	t-value
$\mu_{\max}$	1/h	**0.1335**[a]	11.8	**0.118**	21.0
$X_{\max}$	g/L	**1.154**[a]	17.3	**1.025**	37.9
$Y_{X/C}$	g X/g C	0.484	—	**0.3391**	24.7
Y_{X/CO_2}	g X/g CO_2	0.35	—	**0.3591**	18.9
n_{cap}	–	—	—	**0.3793**	4.3
$Y_{X/N}$	g X/g N	**10.5**[a]	4.9	—	—
k_C	g/L	0.08	—	0.08	—
k_O	g/L	0.001	—	0.001	—
k_d	1/h	0.001	—	0.001	—
Y_{X/O_2}	g X/g O_2	1.352	—	1.352	—
t_{lag}	h	7	—	7	—
m_C	g C/(g X·h)	—	—	0.008	—
m_{cap}	–	—	—	1.5	—
$\rho_{N,\max}$	g N/(g X·h)	—	—	0.5	—
k_{Nup}	g N/L	—	—	0.0001	—
$q_{\min}$	g N/g X	—	—	0.0001	—
$q_{\max}$	g N/g X	—	—	0.096	—
f_{Nrec}	–	—	—	0.5	—
pH_{opt}	–	—	—	7	—
pH_w	–	—	—	10	—
$Y^m_{CO_2}$	g CO_2/g C	—	—	0.05	—
$Y^m_{O_2}$	g O_2/g C	—	—	1	—

Bold: calibrated. [a]From [8] (nutrient-replete calibration dataset), not recalibrated; errors recomputed with the metrics used in this manuscript. —: not in the model or not calibrated. The fixed baseline-only parameters are in Table S1.

III. RESULTS

A. Baseline and Extended Models in Both Conditions

Fig. 1 gives the errors and Fig. 2 the trajectories. The baseline model, calibrated on nutrient-replete data, reached a

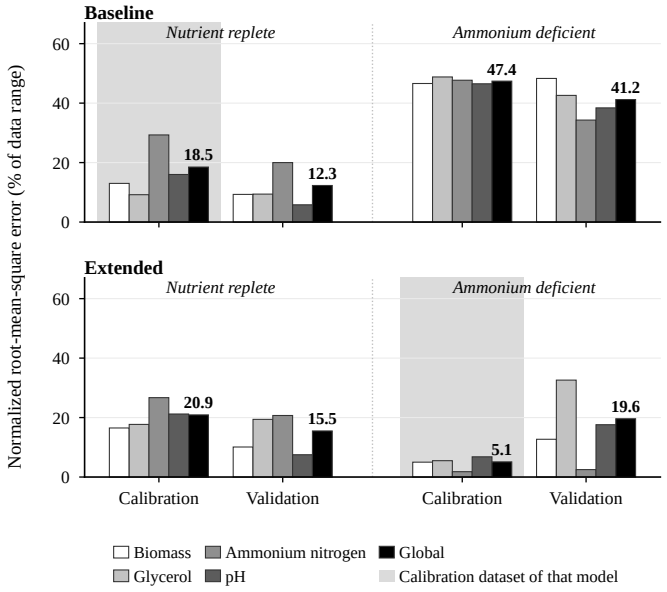


Fig. 1. NRMSE of the baseline and extended models, by variable (biomass, glycerol, ammonium nitrogen, pH, all four pooled), condition and dataset; shaded band: dataset used to calibrate that model.

global NRMSE of 18.5% (calibration) and 12.3% (validation) in the nutrient-replete condition, but 47.4% and 41.2% under ammonium deficiency, where its final biomass in the validation dataset was 0.50 g/L against 1.12 g/L measured. The apparent biomass per nitrogen consumed, $\Delta X/\Delta N$, in the validation datasets was about 12 g/g (nutrient replete) and 28 g/g (ammonium deficient; ratio 2.3), against $Y_{X/N} = 10.5$ g/g in the baseline.

Calibrated on ammonium-deficient data only, the extended model reached 5.1% and 19.6% under ammonium deficiency, -42.3 and -21.6 percentage points relative to the baseline, with all four variables improved on both datasets. Nitrogen was reproduced best (1.8% and 2.5%); on validation the largest errors were glycerol (32.6%) and pH (17.6%), as the model consumes glycerol and acidifies the medium earlier than measured (Fig. 2d). In the nutrient-replete condition, where it was not calibrated, it reached 20.9% and 15.5%, +2.4 and +3.2 points relative to the baseline model, which was calibrated there. The difference is concentrated in glycerol (17.7% and 19.4% against 9.2% and 9.4%), pH and the final biomass: the extended model plateaus at $X_{\max} = 1.025$ g/L whereas the measured final biomass is 1.19 g/L in both nutrient-replete datasets. Nitrogen errors are comparable (26.7% and 20.7% against 29.3% and 20.0%). The internal behaviour of both models (external nitrogen, quota, apparent yields, maintenance share) is described in the Supplementary Material, Sect. S5.

B. Parameter Values, Identifiability and Number of Parameters

Of the 21 parameters of the extended model, 5 were calibrated (16 fixed), against 3 of 14 for the baseline, and 9 of 21 were sensitive (above 0.01). Four sensitive parameters were fixed (t_{lag} and k_O at the baseline values, $pH_{opt} = 7$, $q_{\max} = q(0)$) because, when calibrated, $q_{\max}$ and k_O ended on a bound in most runs, t_{lag} was almost fully correlated with $\mu_{\max}$ and pH_{opt} spanned its whole bound range. The three calibrated parameters of the baseline have t-values between 4.9 and 17.3; the 5 of the extended model between 4.3 (n_{cap}) and 37.9 ($X_{\max}$), 0 below 2, with a largest correlation of 0.94 ($\mu_{\max}$ and n_{cap}; 0 pairs above 0.95). A t-value above 2 is necessary but not sufficient for practical identifiability: $n_{cap} = 0.3793$ has the widest relative error (46%).

Of the initial 552 runs, 120 were admissible (398 had a parameter on a bound) and converge to 11 distinct solutions, whose ammonium-deficient calibration and validation errors have a rank correlation of -0.69 ($p = 0.02$, $n = 11$; -0.65 with the nutrient-replete validation error, $p = 0.03$). A single-parameter fit of $\rho_{N,\max}$ had a worse calibration error (11.0%) than the selected set but lower validation errors (16.7% and 12.6%). Calibrating $q_{\min}$ either sent $X_{\max}$ to its upper bound and raised the nutrient-replete errors to 32.4% and 66.2%, or left n_{cap} on its bound with $t < 2$; with constant maintenance the calibration error was unchanged (5.1% against 5.1%) and the validation error rose to 22.8%.

The final stage compared a nested ladder of four calibration subsets with 4 to 7 calibrated parameters (ten runs each, all admissible; Table S13). Calibrating n_{cap} instead of fixing it at 0.8 lowered $\widetilde{RSS}$ by 20% ($F = 8.9$, $p = 0.005$; ΔAICc$=-6.27$) and the ammonium-deficient validation NRMSE from 22.9% to 19.6%, but the smallest t-value fell from 18.6 to 4.3. Adding $\rho_{N,\max}$ lowered $\widetilde{RSS}$ by only a further 4.7% ($F = 1.7$, $p = 0.21$; ΔAICc$=+1.05$; Akaike weights 0.34 and 0.57 for six and five parameters), left the validation errors unchanged and lowered the smallest t-value to 3.8; adding $q_{\max}$ as well left them unchanged and was disfavored (ΔAICc$=+4.15$ against five parameters), and no other fixed parameter (t_{lag}, m_C, m_{cap}, $q_{\min}$, pH_{opt} with pH_w) gave an admissible run when added.

IV. Discussion

Trade-off between the two models. Reproducing the ammonium effect required adding physics (a state and four further changes), and the extended model gives up 2.4 to 3.2 percentage points in the nutrient-replete condition, where it was not calibrated, in exchange for 42.3 and 21.6 under ammonium deficiency. The comparison is between whole models: besides the quota, the extended model changes the self-inhibition, maintenance, pH factor and charge balance, so part of the improvement may not come from the quota. A fixed yield cannot represent the factor 2.3 in apparent yield between conditions; in the extended model external nitrogen is exhausted at about 18.5 h in the ammonium-deficient calibration dataset and growth continues on the internal pool, the quota falling from 0.096 to 0.055 g N/g X, so it forms 22.1 g of biomass per g of nitrogen consumed, as measured (21.6 g/g), where the baseline allows 10.0 g/g (Supplementary Material, Sect. S5).

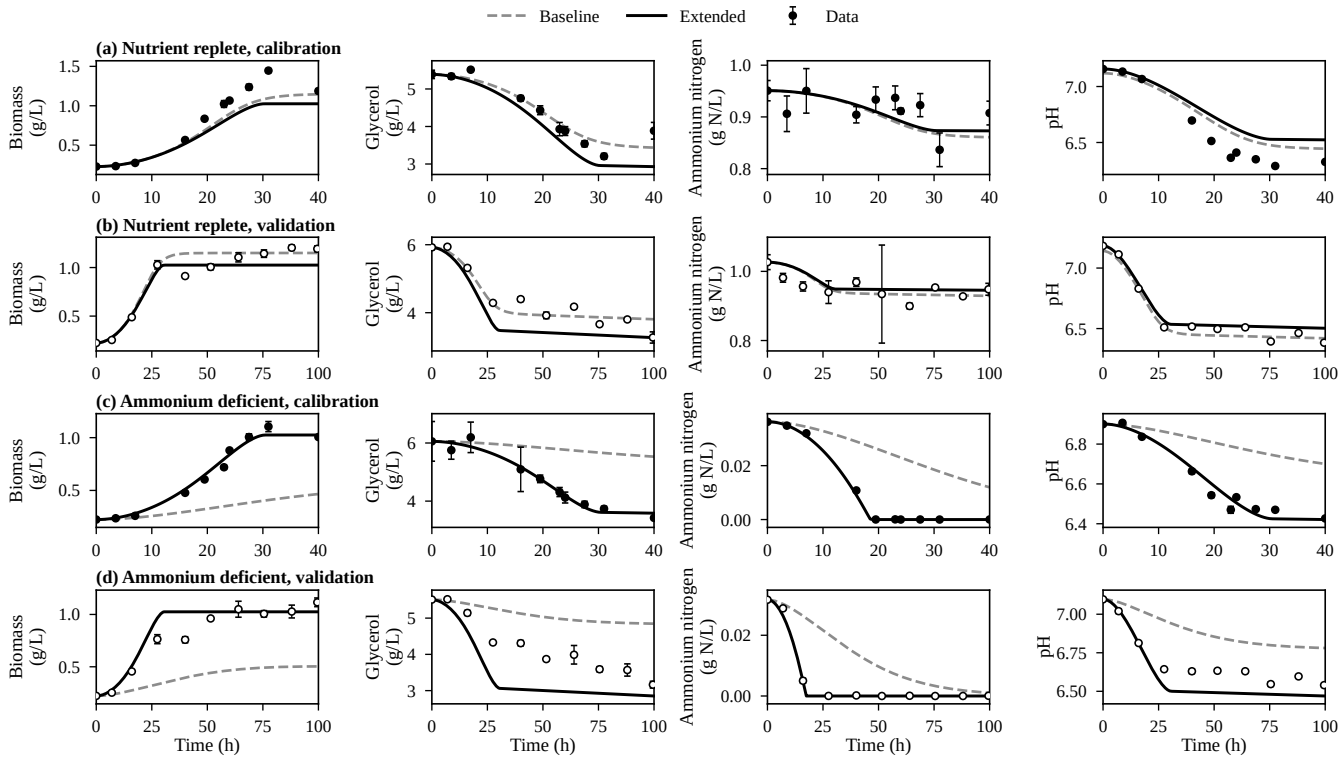


Fig. 2. Calibration and validation trajectories of biomass, glycerol, ammonium nitrogen and pH under nutrient-replete (a, b) and ammonium-deficient (c, d) conditions. Dots: measured means with standard errors (filled: calibration, open: validation). Dashed gray: baseline, calibrated on (a). Solid black: extended model, calibrated on (c).

The quota does not limit growth in the calibrated model. The Droop factor stays at $f_{Nq} \geq 0.998$ in every dataset, because $q_{\min}$ was kept at 10^{-4} g N/g X, far below the quota reached. The plateau, at about $X_{\max}$ (1.025 g/L), is therefore set by the capacity term, in both conditions, and not by nitrogen exhaustion; the quota is a nitrogen pool diluted by growth that decouples nitrogen consumption from biomass formation. Whether nitrogen limitation also contributes cannot be settled with these data, as calibrating $q_{\min}$ was either worse in the nutrient-replete condition or not admissible (Sect. III-B).

Maintenance and the literature. Replacing the biomass-dependent maintenance by a constant worsened the validation error (Sect. III-B), an indication and not a test of its form. Uptake continuing past substrate limitation was reported for *M. smegmatis* [7], which is qualitatively compatible with a growth-independent carbon demand but does not quantify it; applying the same decoupling to nitrogen through the quota is an assumption of this work. Droop kinetics come from phytoplankton ecology, nitrogen-responsive regulation in mycobacteria [5] is only structurally analogous to the uptake feedback of (9) (whose form is also used in [3]), and no measurement of an internal nitrogen pool in mycobacteria is used.

Selection and generalization. The set was selected by calibration criteria (fit, information criteria, F-tests, identifiability), which did not identify the lowest validation error: across the 11 distinct solutions of the first 552 runs, lower calibration error went with higher validation error, consistent with fitting more parameters to ten samples per variable, and the single-parameter fit of $\rho_{N,\max}$ validated better despite a worse calibration error. The differences may be within the noise; the selected set is preferred for fit, parsimony and identifiability, not shown to predict better, and adding $\rho_{N,\max}$ is not excluded by AICc: keeping five parameters is a judgement, not a test result.

Limitations. This is a proof-of-concept comparison on four datasets of ten samples per variable: it does not establish generalization to other batches, strains or scales, and no error has an uncertainty estimate. Four sensitive parameters ($q_{\max}$, t_{lag}, k_O, pH_{opt}) were fixed, so calibrated values depend on them. The charge balance mixes concentration units (phosphate, citrate and CO_2 in g/L, ammonium in mol/L), so the simulated pH is a lumped quantity, not a mechanistic prediction of buffer chemistry (Supplementary Material, Sect. S8). A good fit to four macroscopic variables does not exclude other mechanisms, such as aggregation, which depends on the carbon-to-nitrogen balance [21] and breaks the well-mixed assumption.

V. Conclusions

Calibrated on ammonium-deficient data alone, the extended model removes most of the error that the baseline has under nitrogen limitation: the global NRMSE fell from 47.4% to

5.1% in calibration and from 41.2% to 19.6% in validation (a test without refit). In the nutrient-replete condition, where it was not calibrated, it was only 2.4 and 3.2 points behind the baseline calibrated there: each model is best in the condition it was fitted to, but only the extended model describes both conditions with one parameter set, with a global NRMSE of 20.9% or below on all four datasets. This required calibrating only 5 of 21 parameters, all with t-values above 4 (n_{cap} is the weakest), and a nested comparison under bounds fixed in advance showed no significant gain in fit (F-test) and none in validation from calibrating $\rho_{N,\max}$ or $q_{\max}$. The results support the extended model as a whole, not individual parameter values or specific mechanisms: the improvement is consistent with a quota that acts as a nitrogen pool diluted by growth, although the comparison is between whole models, and the biomass plateau is set by the capacity term, not by nitrogen exhaustion. The extended model is therefore a starting point for a single description of both conditions; intracellular and total nitrogen measurements and an independent ammonium-deficient culture would test it.

Data and Code Availability

The Supplementary Material, code, datasets and calibration outputs are in the GitHub repository *DAE System Model Calibration and Validation* [19].

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